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The Art of Mechamimicry

Designing Prototyping Tools for Human-Robot Interaction

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The Art of Mechamimicry: Designing Prototyping Tools for Human-Robot Interaction

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ABSTRACT

This research investigates the application of tangible and embodied prototyping methods integrated with virtual simulation in Human-Robot Interaction (HRI). We present the development of the "kinematic puppet," a **reliable**, **reusable**, **adaptable,** and **accessible** prototyping tool designed to facilitate stakeholder engagement in early-stage HRI research and development without requiring significant financial or time investments. The potential of this methodological approach is illustrated through a formative co-design workshop in Robotic Assisted Surgery (RAS), where the kinematic puppet, simple props and a low-fidelity anatomical model enabled stakeholders to externalise tacit knowledge through role-play scenarios. The case study suggests that combining physical and virtual approaches can support stakeholders in expressing concrete ideas for improving or changing the interaction, making abstract concepts tangible, with virtual simulation enabling rich data capture for further design development. This work contributes to the rapidly expanding toolbox of design approaches in HRI.

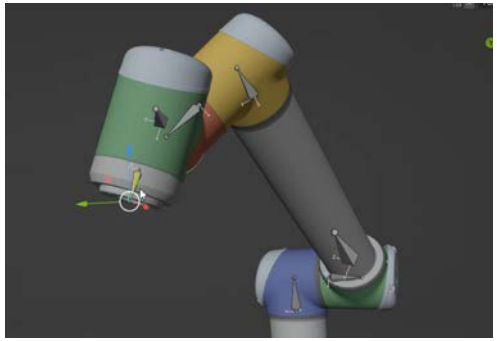
Authors Keywords

Human-Robot Interaction; Prototyping Tools; Design Methods; Tangible Interaction; Embodied Design

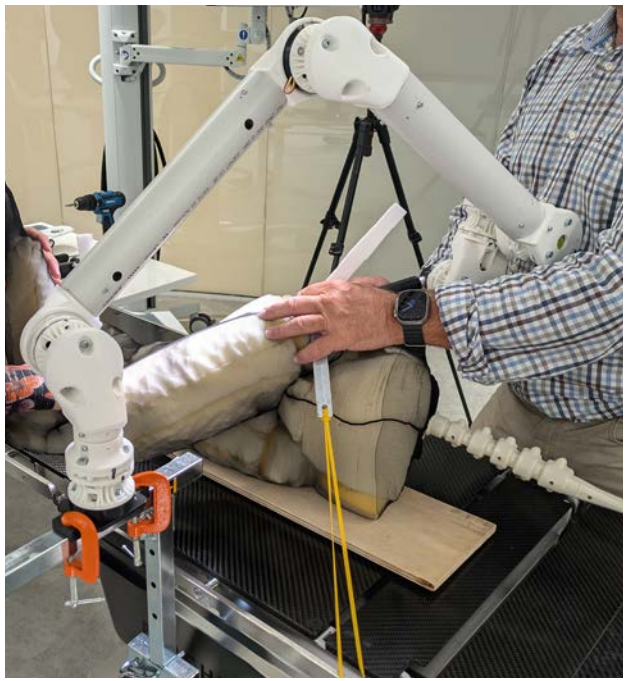
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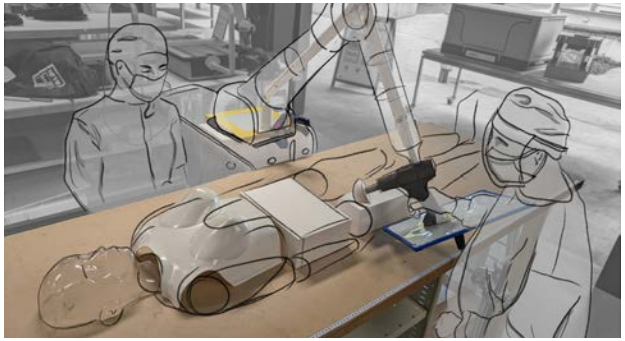
• Human-centered computing ~ Systems and tools for interaction design











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INTRODUCTION

Studies exploring human factors in Robotic Assisted Surgery (RAS) systems illustrate the complex interactions occurring between surgeons, the surgical team, robotic systems, and the broader surgical environment [4, 5, 21]. This includes insights into surgeons' tacit knowledge and adaptive practices during robotic assisted procedures [10, 18]. To create more intuitive and responsive Human-Robot Interactions (HRI) within surgery requires collaborative approaches that bring together clinicians' expertise, roboticists' technical knowledge, and human factors specialists' understanding of human capabilities and limitations [20].

However, effectively integrating these diverse perspectives early in the development process remains challenging. Lupetti et al. [16] argue that design techniques and processes are needed to help bridge the gap between the technical research interests that drive most engineering-based HRI approaches and the actual sociocultural reality and needs of the end-users that these robots will interact with. This disconnect between technical development and user needs is particularly evident in RAS, where there is a need to realign how surgical theatres should operate in theory ('work as imagined') with how they actually function in practice ('work as done') [4, 5].

Early-stage prototyping techniques have the potential to tackle this challenge. Zamfirescu-Pereira et al. [32] emphasise the value of early-stage, low-cost, and flexible exploratory prototyping techniques that elicit rich feedback from end-users. This early engagement in the design process is critical to devising new robot morphologies, actions, and behaviours effectively. Prati et al. [25] suggest that virtual prototyping approaches for human-robot collaboration (HRC) support "solution prototyping" and evaluation with end-users, a critical aspect of early stage design processes. These virtual prototyping approaches support planning and evaluation of interactions with advanced systems in HRI in ways that enable user engagement and reflection [23, 24]. By integrating physical and virtual prototyping approaches, designers can rapidly explore concepts while engaging end-users safely and economically to better support robot adoption [3].

Early stakeholder engagement through development tools and frameworks that enable rapid prototyping and testing is particularly crucial but challenging in high-stakes contexts such as RAS [20]. This context challenges existing design approaches and necessitates a rethinking of prototyping techniques. While virtual environments effectively simulate situated interactions and can support spatial testing and rich data capture, they lack embodiment and haptic feedback [34]; both of which are essential elements of HRI. On the other hand, creating functional HRI prototypes is often costly due to the technical complexities requiring substantial technical expertise and development time, which can hinder rapid ideation and iterative design processes.

Inspired by HRI research utilising roleplaying [15], bodystorming [2, 34], Research-through-Design [6, 17], exploratory prototyping [32], and more playful designerly ways of working, this research investigates the integration of tangible, embodied prototyping with virtual simulation as a hybrid method for early-stage HRI design. It aims to advance HRI methodologies and address the challenges to adoption in high-stakes contexts by enabling more designerly ways of working supported by virtual simulation for rich data capture. This research investigates the question:

How can tangible interfaces integrated with virtual simulation support embodied methods for early-stage HRI prototyping?

This pictorial presents the development of the kinematic puppet, a novel tangible robot prototyping tool, and demonstrates its application in design-led HRI research through the lens of orthopaedic RAS. Used in combination with a low-fidelity anatomical model and simple props, the puppet formed part of a broader prototyping toolkit deployed during a co-design workshop to explore surgeon-robot interactions, workflow spatial layout, tool use, and ergonomic constraints in surgical HRI scenarios.

The kinematic puppet is a passive, articulated, and modular robot skeleton prototyping tool that facilitates HRI exploration by enabling users to puppeteer robot movements and behaviours through a Wizard-of-Oz (WoZ) control approach. Constructed using 3D printed joints with embedded magnetic rotary encoders and connected by PVC pipe linkages, its adaptable design supports the rapid assembly of different robot arm morphologies. By providing a physical anchor, the kinematic puppet enhances virtual simulation interactions and supports embodied prototyping of robot movements and behaviours. Similar to Zhang et al. [34] 'Wizard of Props' approach, this allows users to role-play scenarios within adaptable environments, making the exploration of robot interactions accessible to those without extensive technical expertise or resources.

In this pictorial, we present three contributions to HRI research. First, the kinematic puppet, a tangible, modular prototyping tool that enables rapid construction and manipulation of reconfigurable robot arm forms, allowing for embodied and speculative exploration of human-robot interactions without requiring programmed or actuated systems. Second, a hybrid prototyping method that combines physical puppeteering, roleplaying, and virtual simulation to support the elicitation of tacit knowledge and the generation of latent knowledge through enactment, enabling participants to explore and articulate novel applications of robotic systems within complex, domain-specific tasks. Third, a design-led approach to support interdisciplinary collaboration in the early stages of HRI prototyping within high-stakes, technically constrained domains such as surgery, where such practices remain underexplored.

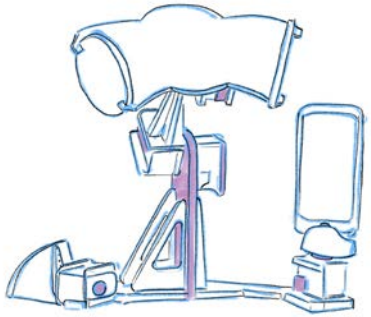
BACKGROUND



Kelley [12]

Wizard-of-Oz (WoZ) Prototyping

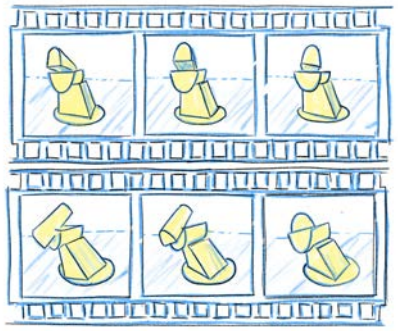
Is a common approach to establish the basis for a proof-of-concept evaluation. This technique involves a human controlling the behaviour of the robot system, which is commonly done without the participant's awareness of the control method [8]. These control methods range from full teleoperation to partial control of 'higher-level' decision-making processes [7].



Hoffman & Ju [9]

Skeleton Prototyping

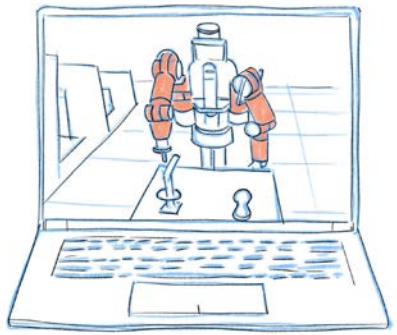
Is used to explore robot form and movement through the creation of a "full robot" skeleton model. These prototypes can be either passive models that do not incorporate movement while still allowing for the examination of form and ergonomics or active models that utilise actuators to give the prototype dynamic functionalities. This approach commonly uses rapid prototyping techniques such as laser cutting and 3D printing to create the robot's form, allowing the robot to be assembled in a matter of days with little waste in material [9].



Hoffman & Ju [9]

Video Prototyping

Extends on the previous approaches and uses recordings of physical movement performance of actuated skeleton movement, rendered clips from 3D animation studies, or even abstract shapes [9]. This method ensures that all participants can view the exact same movements allowing for greater experimental control [17]. Hoffman & Ju [9] propose that the addition of video to the improvisational aspects of the other methods discussed facilitates a more collaborative design approach.



S. Zhang et al. [33]

Virtual Simulation Prototyping

Positioned between the prior approaches, it provides a balance of immersion and practicality. While development of virtual simulations may require more time and resources than video prototypes, it offers superior immersion by enabling real-time observation and documentation of participant interactions. For example, pre-programmed motions can elicit reactions to intentionally designed cues [26], while integrating human-in-the-loop control or WoZ techniques allows capturing responses to spontaneous and adaptively timed cues [32].

Emerging Technologies

Sosa-Ceron et al. [28] advocate for the use of emerging technologies such as Augmented Reality (AR), Virtual Reality (VR), and Digital Twins (DT) within HRI design and research to further enhance the safety and efficacy of testing environments while also reducing ergonomic strain and fostering more natural HRI. The application of DT, in particular, has shown promising results in industrial settings, not only improving safety and interaction but also reducing both implementation time and costs [26]. Building on these developments, this work adapts established prototyping techniques such as WoZ, bodystorming, and skeleton prototyping for contexts like RAS, where such design practices are underexplored. While puppeteering in HRI has been used to explore expressive movements and gesture-based interactions, the kinematic puppet is designed for task-oriented interaction in constrained environments, grounded in the movement logic of robot arms. While the system supports VR and AR, in this study virtual simulation was used primarily to record and replay movement sequences to support reflection, analysis, and further development. This focus responded to input from the surgical participant, who preferred physical anatomical models, prompting us to prioritise tangible representations over immersive environments during this phase.

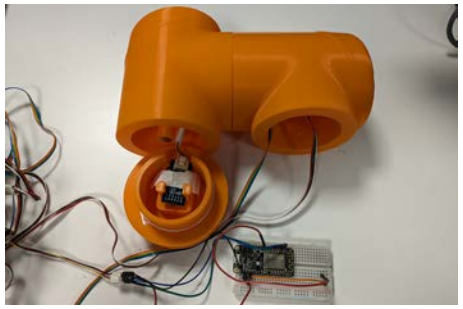
DESIGN RESEARCH METHODOLOGY

This research is structured across four design phases, each contributing to the development of a tangible and adaptable prototyping tool to support early-stage HRI design. The development and use of the tool were shaped by ongoing collaboration with an industry partner specialising in surgical robotics. Taking a research-through-design approach, the development process positioned prototyping as a means of inquiry, iteratively generating and testing artefacts that could elicit embodied knowledge, surface technical constraints, and support reflection on possible futures. Rather than progressing toward a fixed solution, the aim was to develop a prototyping tool that could facilitate HRI exploration and interdisciplinary dialogue within constrained, high-stakes domains such as RAS.



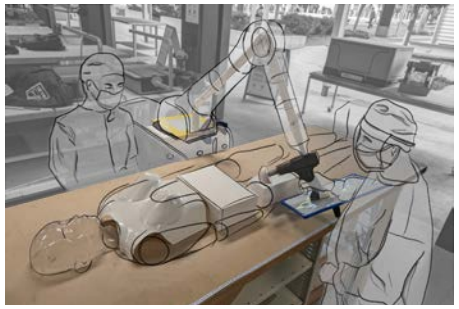
Phase One: Understanding Through Comparative Design Experiments

Gaining familiarity with robotic systems through simulation platforms, tangible programming methods, and skeleton prototyping, using the UR10 as a benchmark to explore robotic movement and form.



Phase Two: Developing a Tangible Interface Integrated with Virtual Simulation

Exploring physical manipulation as a means of intuitive robot control, while developing virtual simulation integration and data capture methods to support later analysis, reflection, and replication of movement behaviours.



Phase Three: Contextualising Through Scenario Reconstruction

Reverse engineering a surgical HRI scenario (total hip arthroplasty) to examine interaction, spatial layout, and tool design using roleplay and bodystorming methods, without the constraints of a specific robotic platform.



Phase Four: The "kinematic puppet" A Modular & Adaptable Prototyping Tool

Refining the kinematic puppet as a modular, sensor-enabled prototyping system capable of controlling a digital twin. This phase culminated in a Research-through-Design study with a surgeon to explore the puppet's application in surgical HRI tasks.

Design Priorities Shaping Approach

Accessibility: lowering technical barriers to engagement by enabling use without programming or reliance on a specific robotic platform.

Task Relevance: maintaining fidelity to the movement logic and constraints of real robot arms to support the exploration of spatial layout, workflow, and ergonomics.

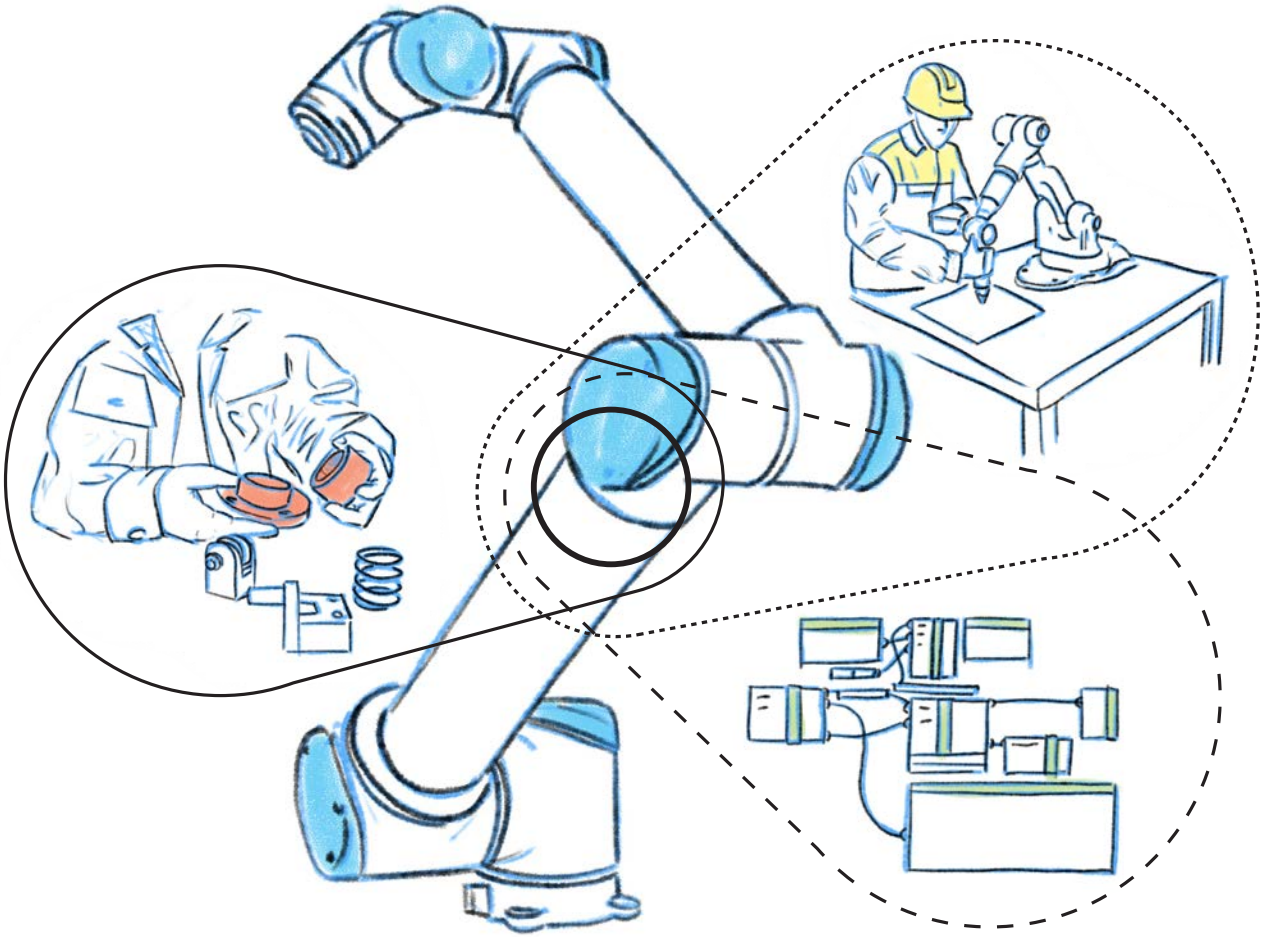
Modularity: allowing for reconfiguration to test different robot morphologies and interaction scenarios.

Formative Case Study: Applying the Toolkit in surgical HRI

Following this development process, the toolkit was applied in a co-design workshop involving a cross-disciplinary working group. The 90-minute session included a practising orthopaedic surgeon, their assistant, a robotics engineer from the industry partner organisation, and three members of the design research team. The workshop served as a formative case study, demonstrating how the kinematic puppet and associated design methods could support stakeholder engagement in a highly constrained, high-risk context such as RAS. Rather than seeking generalisable findings, the aim was to explore how techniques such as bodystorming, roleplay, and embodied prototyping might facilitate collaborative ideation and scenario development within a domain shaped by technical constraints, ergonomic considerations, and clinical expertise. Data from the session were captured through design artefacts, sensor data, observational notes and video recordings and were analysed using a combination of reflective analysis and video analysis.

PHASE ONE: UNDERSTANDING THROUGH COMPARATIVE DESIGN EXPERIMENTS

Coming from a design and psychology background with limited robotics experience, it was important to first develop foundational knowledge of robotic systems to inform the prototyping tools and approaches. The initial phase focused on exploring accessible approaches to robot programming and prototyping, investigating three key areas: simulation-based programming tools suitable for non-technical users, tangible programming methods like lead-through programming, and skeleton prototyping techniques as a foundation for physical robot design exploration. We grounded this investigation using the UR10[1] robotic arm, chosen for its widespread application in research and industry and its accessibility through simulation platforms like URSim and Grasshopper's Robots plugin. Informed by skeleton prototyping approaches, we reconstructed a low-fidelity model of the [1]UR10 [31] to understand its form and kinematic structure. Replicating an existing system provided a clear benchmark for understanding basic robotic principles and prototyping challenges.



[1]UR10 (Universal Robots A/S, Odense, Denmark).

Rapid Physical Prototyping

Rapid prototyping methods including cardboard and foam modelling were used to create a low-fidelity robot model. While cost-effective and quick to produce, these materials proved too fragile for repeated handling and too compliant to accurately replicate robot movement. These challenges highlighted the need for more robust materials that would be **reliable** and **reusable** to support repeated use as a prototyping tool.

Lead-through Programming

This approach provided valuable insights into the benefits of being able to physically manipulate the robot as a method of programming. However the requirement for a real robot is costly and not always **accessible** for early-stage prototyping. It also requires a commitment to a specific platform, making it impractical to explore multiple robot options simultaneously, limiting **adaptability** in early-stage design and development.

Simulation and Visualisation

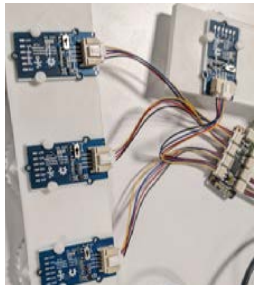
These visual programming methods, particularly in the case of the Grasshopper platform which uses nodal programming, were quite **accessible** for a designer and could be run from most laptops. However, these methods still require a level of technical knowledge and skill development in the use of the software, access to a software license and some understanding of the principles of programming.

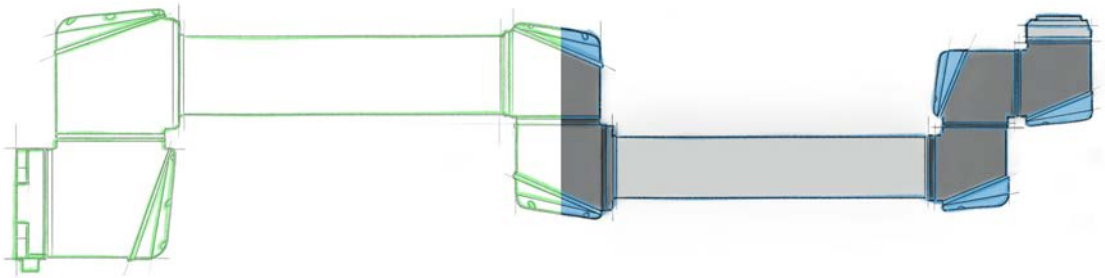
PHASE TWO: DEVELOPING A TANGIBLE INTERFACE INTEGRATED WITH VIRTUAL SIMULATION

Reflecting on the Phase One design experiments, Phase Two focused on developing a prototyping tool that combined dynamic virtual robot manipulation with the tangibility of lead-through programming. From these reflections emerged the concept of a tangible interface integrated with virtual simulation. This direction was inspired by the "Dinosaur Input Devices" (DIDs) used during the production of Jurassic Park [14], these encoder-fitted armatures enabled stop-motion animators to intuitively control complex digital models using their existing embodied expertise. The aim was to create a tangible interface for early-stage robot prototyping that retained the **accessibility** of physical manipulation while leveraging virtual simulation for analysis and iterative refinement. To improve the **reliability** and compliance of the model, fabrication shifted to 3D printing. While the prototype continued to replicate the UR10 as a benchmark for robotic form, this phase introduced embedded sensors to capture the puppet's movements for replication within simulation.

Virtual Design Experiments

Integrating Sensor Data Using Arduino

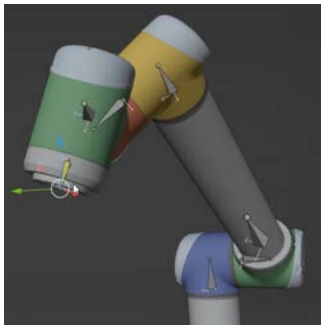




Joint Velocity Control

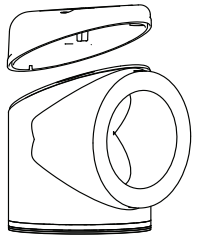


Inverse Kinematic Control

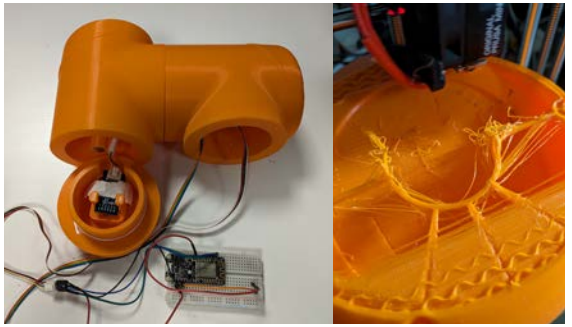


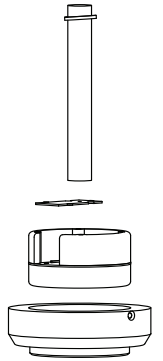
Physical Design Experiments

Rotary Encoder Mount Design



3D Printing for Rapid Prototyping





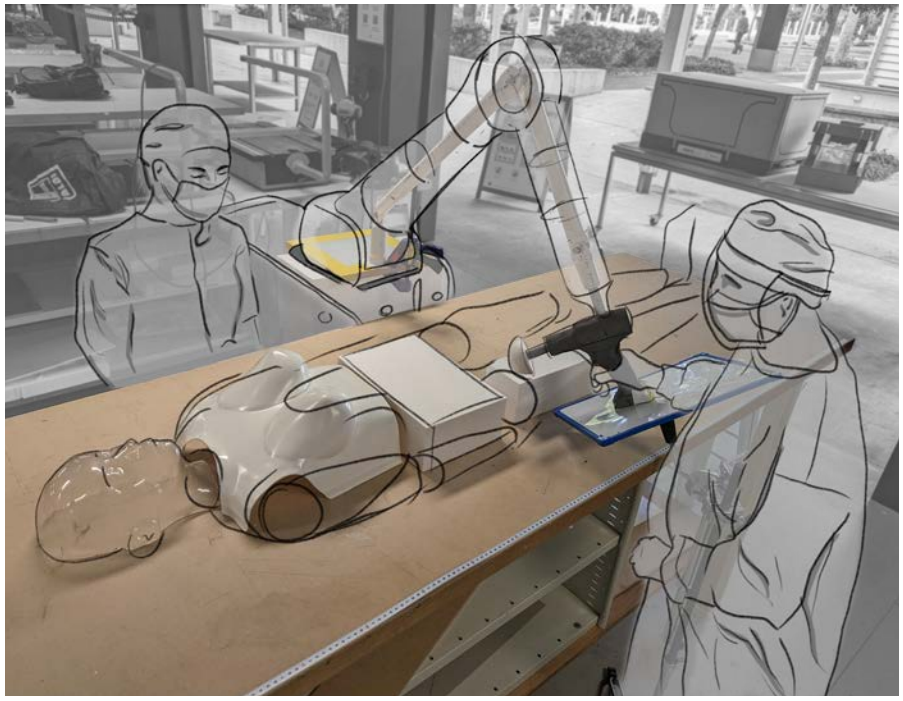
Two methods of robot position capture were explored: using rotary encoders to directly translate joint movements into simulation, and attaching a VR controller to the physical model's end-effector and using a 3D model rigged for inverse kinematics, though this proved technically complex to implement **reliably**. We chose direct translation as this would be **adaptable** across different robot platforms and allow users to physically experience joint limits, mechanical constraints, and kinematic chain effects. This approach better supported user understanding of robotic movement in HRI prototyping scenarios.

The 3D printed UR10 model was fitted with magnetic rotary encoders which would allow for the direct translation of the robot models movements within simulation. While the 3D printed models were more **reliable** and accurate than foam and cardboard, they were still relatively costly, with prolonged print times that were prone to failure. It was decided that printing an entire shell for each robot to explore different robotic platforms within early-stage exploratory design processes would be expensive and inefficient as they would not be **reusable** if prototyping explorations shifted to a different platform.

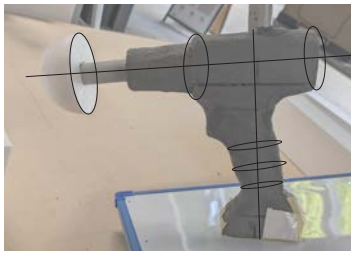
PHASE THREE: CONTEXTUALISING THROUGH SCENARIO RECONSTRUCTION

We decided to re-evaluate the underlying aim of the project and what these rapid prototyping approaches could provide: tangible and embodied ways of exploring HRI concepts through role-playing, bodystorming, and WoZ prototyping methods supported by virtual simulation. With this in mind, we decided to explore an HRI exemplar. This reverse engineering approach aimed to understand how human-robot interactions could be replicated using tangible rapid prototyping, with a focus on role-playing and bodystorming methods, without having to commit to a specific robotic platform. A Total-Hip Arthroplasty (THA) using RAS was chosen as it could be studied through direct observations and publicly available video sources. This procedure uses computer-controlled robotic systems to assist surgeons in precisely planning and executing the replacement of the hip joint and femoral head with an implant [8]. This allowed for the exploration of various aspects of human-robot interaction, including robot morphology, spatial layout, and end-effector tool designs, each relevant to different tasks within a larger process.

Anterior Approach to RAS based THA

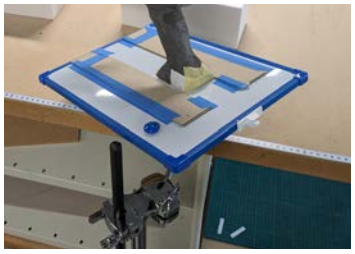


Tool Prototyping



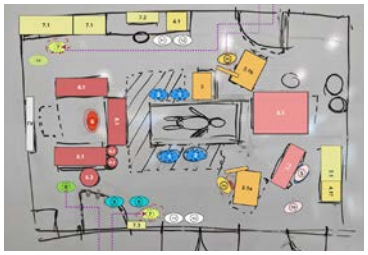
Foam and cardboard were used to rapidly prototype surgical tools and cutting attachments. These components were fitted with magnets to adhere them to the physical planar surface.

Haptic Feedback Prototyping



We discovered the significance of different forms of haptic feedback within RAS systems. For computer assisted orthopaedic surgery (CAOS) these include virtual 'rigid planar surfaces' that create haptic boundaries, assisting the surgeon to control the tool's movement during the surgical procedure. These boundaries effectively reduce degrees of freedom (DOF) that need to be controlled simultaneously by the surgeon, aiding in executing precise movements with enhanced stability [1, 22]. These observations inspired the concept of physical analogs to replicate these complex programmatic control systems, using a magnetic surface fitted to an adjustable stand. This approach provided a more intuitive and **accessible** way to understand and explore these robotic control parameters in a tangible way.

Layout Analysis



Direct observations and publicly available video sources were used to understand the environmental layout and team dynamics occuring within the operating room. These are highly dynamic environments with limited space for additional equipment.

Rapid Joint Design



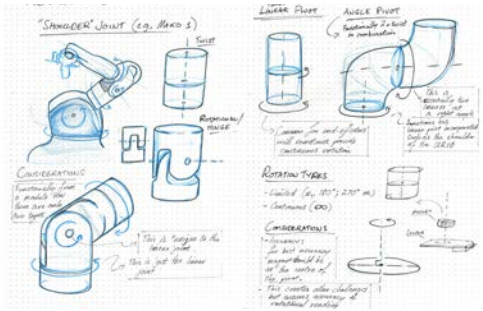
A simple kinematic model of a novel robot design was created using PVC pipe with joints made from bolts and washers. Using this **adaptable** setup, different parts of the interaction such as acetabular reaming could be played out using bodystorming approaches.

PHASE FOUR: THE “KINEMATIC PUPPET” A MODULAR AND ADAPTABLE PROTOTYPING TOOL

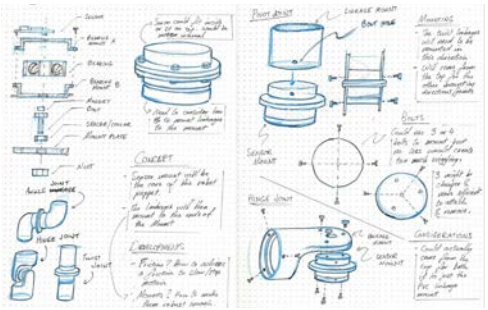
Learning from the earlier phases we sought to create a **reliable**, **reusable**, **adaptable** and **accessible** robot prototyping tool with the ability to control a digital twin in simulation. Our design challenge encompassed three key requirements: develop components that could replicate a variety of robot arms, house the required electronic components for capturing joint positions, maintain a low weight for ease of use while ensuring the parts were robust enough for repeated use. Through this design process we discovered the key to modularity was the idea of a core component. A rotational housing for sensors that different ‘Joint Mount’ components could be fitted to provide linear, angled, and hinge-based rotations. PVC pipe could then be cut to required linkage lengths keeping the cost relatively low. This system enabled rapid reconfiguration of kinematic arrangements significantly expanding the prototyping tools adaptability compared to earlier iterations. To validate this approach and begin exploring its potential for HRI studies, a kinematic puppet based on the MAKO™[2] system [29] was built. This model was used as part a Research-through-Design study with a surgeon to discuss the practical applications of this new tool in HRI prototyping for RAS in orthopaedic applications.

Abstracting Robots

Robot Design Analysis



Component Design

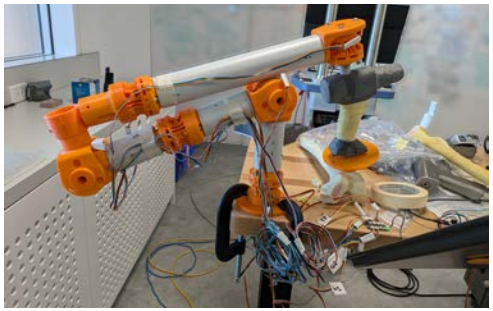


Development and Testing

Component Testing and Re-Design



Full Build & Testing

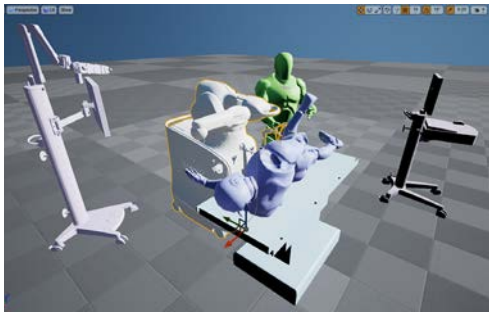


Research-through-Design Session

Physical Prototype



Virtual Simulation



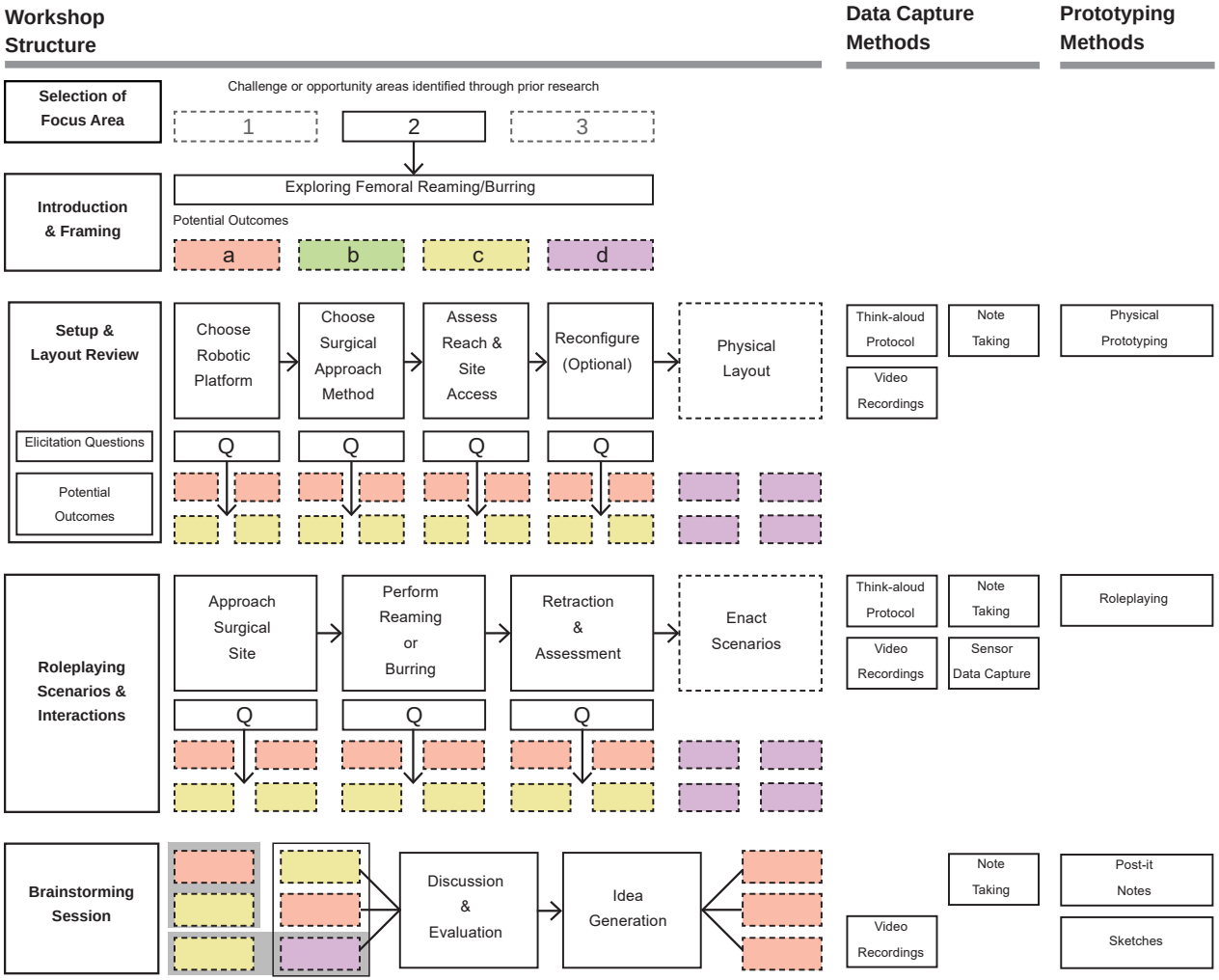
Session Outcomes

Initial plans for the co-design workshop centered on utilising extended reality (XR) for surgeons to interact with a virtual patient anatomy as this was felt to be a more flexible approach. However, through stakeholder feedback during preliminary sessions, it became evident that physical representation of patient anatomy was crucial for effective prototyping activities, as spatial and tactile understanding of anatomical structures fundamentally informed surgical decision-making. This insight led to a methodological pivot, shifting emphasis toward physical prototyping elements while leveraging virtual simulation primarily for capturing and analysing the kinematic puppet’s movements during role-play scenarios. This adaptation exemplifies how early stakeholder engagement can critically inform prototyping methodology development.

[2]MAKO SmartRobotics™ (Stryker Corp., Kalamazoo, MI, USA).

CO-DESIGN WORKSHOP STRUCTURE

The workshop was designed to explore how HRC could be applied to femoral preparation during revision THA surgery, where an exisiting hip implant is removed and cement is cleared from the femoral canal to accommodate a new implant [13]. The workshop was divided into three stages. Stage 1: Setup and Layout Review involved defining the surgical environment, necessary tools, and workflow. Stage 2: Roleplaying Scenario and Interactions allowed participants to engage hands-on with the kinematic puppet, providing real-time feedback. Stage 3: Brainstorming synthesised insights and generated ideas for further refinement.



This workshop serves as an example of how both the kinematic puppet and workshop structure can be adapted to different HRI contexts through context-specific preparation of the prototyping toolkit to enable meaningful stakeholder engagement.

Context of Exploration

The task-specific approach, in this context, helped to mitigate one of the fundamental problems encountered in exploratory research. Sanders & Stappers [27] describe this as the “chicken and egg problem.” By focusing on a defined task the workshop provides a structured approach to planning activities and offers some predictability around potential outcomes.

Participant Considerations

Some participants may be sceptical about unfamiliar co-design methodologies, but an approach that accommodates “sceptics” should also work for “advocates” [27]. Allowing participants to “set the scene” through the setup and layout activity can support buy-in. Once they are comfortable and see value in the process, the conversation can expand to more open-ended or systemic issues.

Session Structure and Workflow

The proposed study had inherent limitations in terms of participants time and availability. The short focused session was a practical way to manage these constraints while still generating meaningful, contextually relevant insights. It provided a focused entry point to identify underlying assumptions, which can be explored further in future research.

Framing Prototyping and Design Process

Framing prototyping activities as opportunities for critical reflection and exploration, rather than as steps toward a final design, can help prevent participants from defaulting to a solution-focused mindset. This aligns with iterative design principles, where insights gained through prototyping prompt reconsideration and refinement of assumptions.

CO-DESIGN WORKSHOP SESSION: EXPLORING HRI IN REVISION TOTAL HIP ATHROPLASTY

Constructing Prototyping Toolkit

Building Kinematic Puppet



Building Anatomical Model



Mock Operating Room Setup

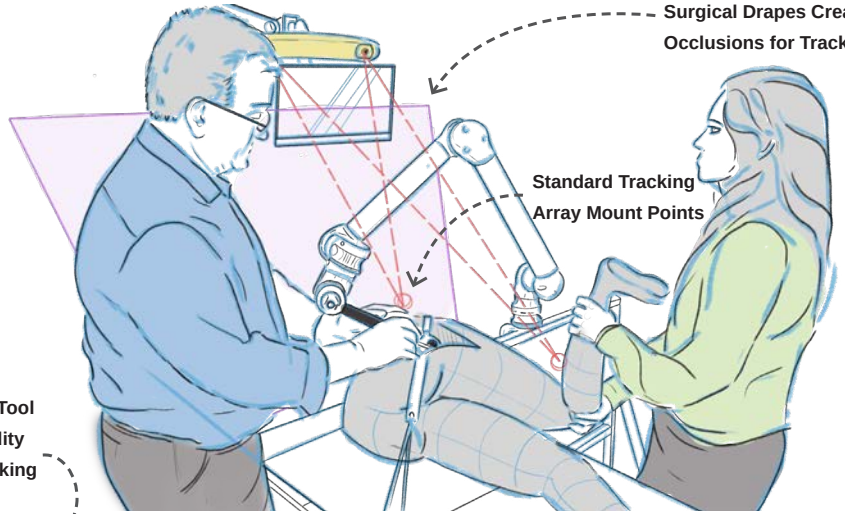


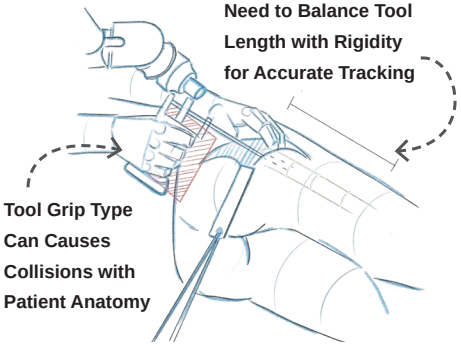
A kinematic puppet of a UR5[3] [30], on an adjustable stand was created for the co-design workshop, chosen for its novel structure compared to current orthopaedic robotic systems. A low-fidelity anatomical model was constructed from a dollar store skeleton modified for anatomical accuracy, and to fit a 3D printed femoral stem before being covered in shaped memory foam. The prototyping toolkit included supporting props including cutting tools, robot handles, a prosthetic implant and soft tissue retractors.

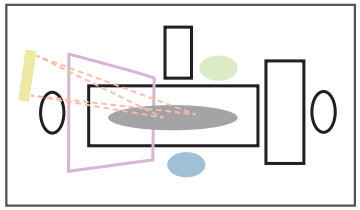
[3]UR5 (Universal Robots A/S, Odense, Denmark).

Co-Design Workshop Session Outcomes

A number of different setups and layout configurations were explored. The configuration shown here is an example of how these low-fidelity prototyping sessions using roleplay and bodystorming can inform decisions ranging from ergonomics to system architecture, including critical requirements for tracking system placement and field of view coverage in computer-assisted orthopedic surgery.

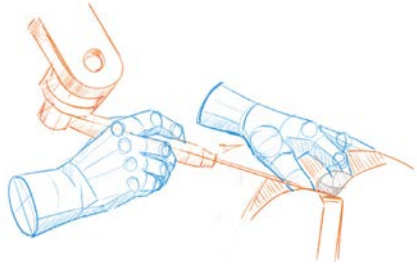




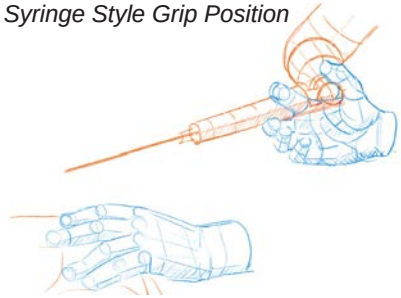


It was found that one of the more ergonomically comfortable configurations would require the tracking camera to be placed towards the patients head in the area normally reserved for the anaesthetist, with possible occlusion of the tracking arrays from the surgical drape.

Pen Style Grip Position



Syringe Style Grip Position



Underhand Style Grip Positon



Two grip types were explored that represented the ergonomic forms of common cutting tools: a drill-type grip and a dremel/pen-type grip. The surgeon explored different grip positions with these tools. While the pen-style grip provided better fine motor control and access than the drill tool, the surgeon noted that the level of force required over the course of the surgical task would be strenuous on the fingers, potentially causing repetitive strain injuries long-term.

AN APPROACH FOR HUMAN-CENTERED ROBOT DESIGN USING A KINEMATIC PUPPET

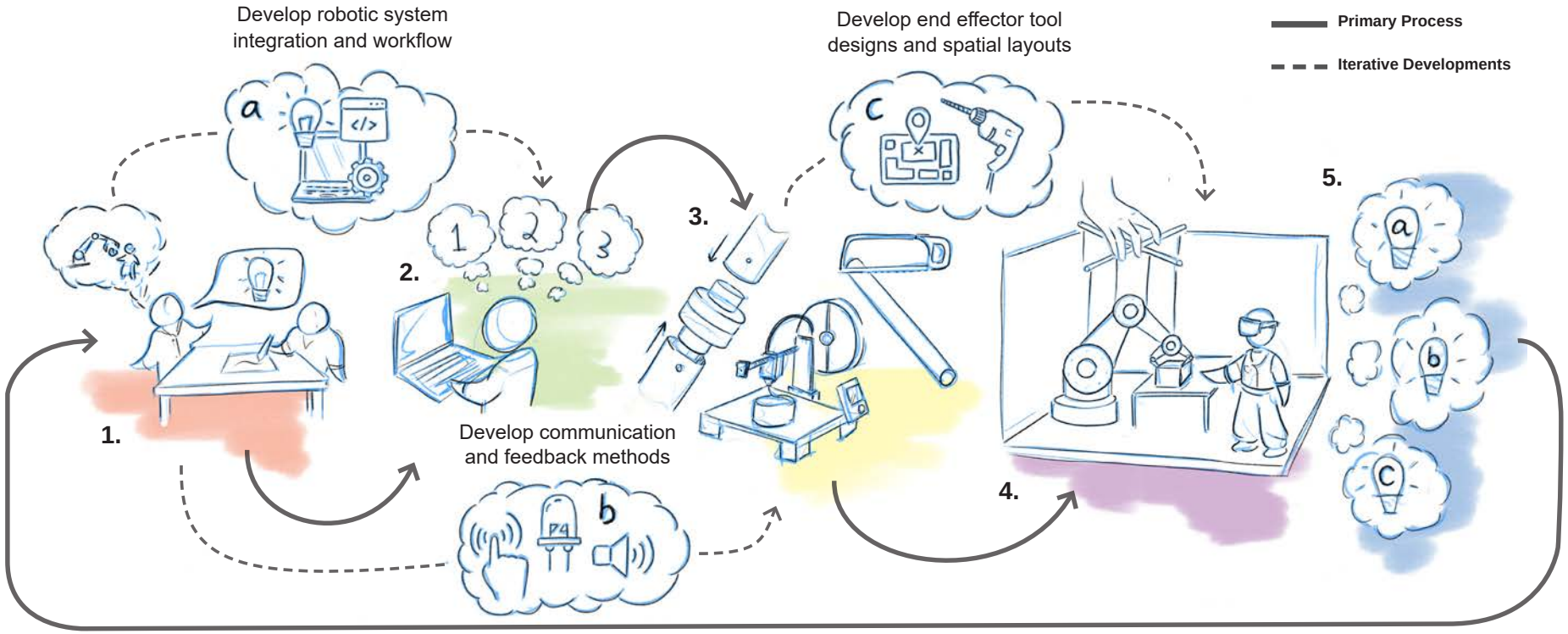
1. Discovery Workshop: Collaborate with users and stakeholders to understand current challenges and opportunities.

2. Analyse Findings: Analyse findings to understand the key stakeholder needs, technical requirements, and possible interaction scenarios that will drive the workshop session.

3. Construct Prototyping Toolkit: Build a kinematic puppet and suitable supporting props and prototyping elements based on the workshop session requirements.

4. Co-Design Workshop Sessions: Use roleplay, bodystorming, and WoZ methods in structured sessions to identify opportunities and challenges through experiential insights.

5. Report and Refine: Document and report findings and refine concepts and ideas for further development.



The approach shown describes a process of repeated engagement with stakeholders and end-users to develop HRI within a given area of application. The approach is intended to be flexible, adaptable and iterative with a focus on engagement with end-users early in the process to understand the challenges and opportunities present within the chosen context. Through structured yet adaptable activities, the approach supports progression from initial context understanding through to concrete prototyping activities, maintaining stakeholder involvement throughout. Each stage builds upon insights from the previous ones while allowing for iteration and refinement based on emerging findings. This approach ensures that technical development remains grounded in real user needs while providing clear documentation of requirements and constraints for further development. The approach can be implemented across different application domains while maintaining its core focus on human-centered design principles and meaningful stakeholder participation.

REFLECTIONS AND CONCLUSION

This pictorial presents the development and application of the kinematic puppet as a tangible, embodied tool for early-stage HRI prototyping. Designed for constrained, task-oriented domains like surgery, the kinematic puppet enabled participants to physically explore robot movement, spatial layout, tool use, and ergonomics. By engaging in embodied enactments of speculative surgical scenarios, we were able to surface surgeons' tacit knowledge that would otherwise be difficult to verbalise. Virtual simulation extended this process by creating a replayable 3D kinematic recording of robot movements, enabling post-session reflection and analysis beyond what observation or video recordings could offer.

Within the workshop, the kinematic puppet and the supporting design methods acted as more than representational stand-ins. Used alongside simple props and an anatomical model, the puppet supported situated roleplay. This enabled participants to investigate existing practices and reflect on possible futures based on past experiences [27]. These design activities revealed not just preferences or ergonomic constraints, but also informed ideas about system configuration, surgeon–robot positioning, and possible cooperative control methods. These design insights would have been difficult to generate through verbal discussion alone.

The workshop process aligned closely with Matthews and Wensveen's framework for understanding prototypes in design research [19]. The puppet served as both a means of inquiry (Role 2), enabling the collection and recording of embodied movement patterns, and a vehicle for inquiry (Role 4), where the prototyping process throughout the workshop became a method of generating knowledge through iteration. This process helped generate latent knowledge and made abstract concepts, such as the impact of robots on surgical workflow and cooperative tool use more tangible. It highlighted the epistemological role of tangible, embodied prototyping methods in uncovering and testing ideas in situ. Virtual simulation supported both roles by allowing precise movement data to be recorded and replayed for further interpretation. This integration of tangible interaction with digital capture positioned the kinematic puppet as a hybrid tool that bridges conceptual design with later technical development.

The artefacts created and modified during the workshop, which included an anatomical model, adapted props, kinematic puppet configurations, and refined spatial setups, functioned as epistemic objects [11]. Through collaborative prototyping, these design artefacts not only supported discussion but also captured evolving knowledge and acted as speculative proposals for possible and desired futures in Robot-Assisted Surgery. Rather than offering fixed solutions, they raised new questions: How does the robot's position relative to the surgeon affect workflow and ergonomics? How do tracking systems constrain layout? What level of autonomy or control is desirable? These provocations facilitated reflection, revealed constraints, and opened space for ongoing inquiry.

Beyond the puppet itself, the workshop served as an example for design-led interdisciplinary collaboration in highly constrained environments. By engaging surgeons, engineers, and designers in a collaborative, hybrid design process that integrated physical prototyping with virtual simulation, this research demonstrated how designerly approaches can mediate between embodied knowledge and technical systems development. The designer's role as translator and facilitator was critical to bridging these disciplinary perspectives and generating insights that may not emerge in siloed research settings.

While this work was situated within robotic-assisted surgery, the kinematic puppet, the hybrid prototyping method and design-led collaborative approach developed through this research have broader relevance. The ability to rapidly construct, enact, and capture human–robot interaction scenarios without relying on fully operational systems reduces the technical threshold for participation. This opens pathways for applying embodied, speculative design methods to other high-stakes, high-constraint domains, such as manufacturing, logistics, or aerospace, where human expertise, spatial constraints, and interaction workflows must be understood early in the design process.

In doing so, this research contributes to a growing toolkit of design-led methods in HRI. It demonstrates how **reliable**, **reusable**, **adaptable**, and **accessible** prototyping tools, when coupled with reflective, interdisciplinary design practices, can support the development of robotic systems that are more closely aligned with the needs, practices, and insights of those who will ultimately work with them.

ACKNOWLEDGEMENTS

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